

THETA-H CONTROLLER ARCHITECTURE

$$u(t) = -K * x(t) + u_{\text{theta}}(t) + u_{\text{BK}}(t)$$

where:

$$u_{\text{theta}}(t) = K_p * e(t) + K_i * \text{integral}(e \, dt) + K_d * \dot{e}(t)$$

$$u_{\text{BK}}(t) = \alpha * u_{\text{BK}}(t-dt) + (1-\alpha) * u_{\text{fb}}(t)$$

with $\alpha = 0.98$ (memory factor)

Parameters used:

$$\tau_1 = 0.5, \tau_2 = 0.4, \tau_3 = 0.1, K_1 = 8, K_2 = 6$$